Ex No: 3a – Blocks World Program

Program

```
class BlocksWorld:
    def __init__(self):
        self.state = {
            "A": "B",
            "B": "table",
            "C": "table"
        }
        self.goal = {
            "A": "B",
            "B": "C",
            "C": "table"
        }

    def is_goal_state(self):
        return self.state == self.goal

    def move(self, block, destination):
        if block in self.state and self.state[block] != destination:
            print(f"Moving {block} from {self.state[block]} to {destination}")
            self.state[block] = destination

    def plan_moves(self):
        print("\nInitial State:", self.state)
        while not self.is_goal_state():
            for block, target in self.goal.items():
                if self.state[block] != target:
                    self.move(block, target)

        print("\nFinal Goal State Reached:", self.state)

bw = BlocksWorld()
bw.plan_moves()
```

Output

```
Initial State: {'A': 'B', 'B': 'table', 'C': 'table'}

Moving B from table to C
Moving A from B to B
Moving C from table to table

Final Goal State Reached: {'A': 'B', 'B': 'C', 'C': 'table'}
```